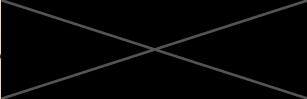


National University of Computer and Emerging Sciences (FAST)

Applied Calculus(EE)

Mid-I (Fall' 2015)

Roll No: 

Section: 

Time allowed: 60min

Max.Marks:40

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions, all questions carry equal marks.
2. Do not ask any question about the contents of this examination from anyone.
 - a. If you think that there is something wrong with any of the questions attempt it to the best of your understanding.
 - b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
 - c. Write all steps, missing steps may lead to deduction of marks.

Q1. Solve the following inequality

$$\frac{x-2}{x+3} \leq 1$$

Q2. Find the following limit or explain why the limit does not exist?

$$\lim_{x \rightarrow -1} \frac{2}{(x+1)} |x+1|$$

Is the function continuous at $x=-1$?

Q3. Evaluate $\lim_{x \rightarrow 2} \frac{1}{(x-2)}$

Also find asymptotes for the curve.

Q4. Graph the function, not by plotting points, but starting with the graph of one of the standard functions and applying the appropriate transformations.

$$y = 1 - 2\sqrt{x+3}$$